

MySQL and Its Domination in the World of Databases

Data has evolved as one of the most important commodities for any company today. Read on to understand how MySQL has emerged as a trusted and reliable database option.

MySQL appears to be one of the most extensively used databases available today.

Even though other open source databases have grown over the years, they still trail MySQL in terms of popularity. MySQL is not only the most liked database among users, but is also the preferred database for developers as surveyed by Stack Overflow and JetBrains. It has managed to become their database of choice and the one that they want to continue to stick to as well. All the major social, e-commerce, tech, finance and manufacturing companies are relying on this database today.

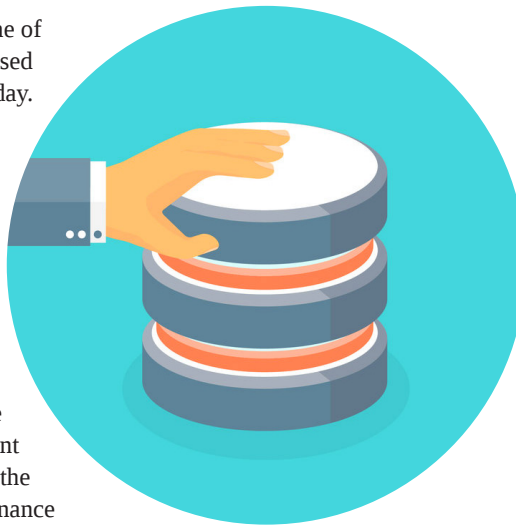
What makes MySQL so popular?

It gives customers what they want:

- MySQL has kept up with current trends while making sure things are not complicated for customers.
- It has kept things simple and easy. It never lets its customers down:
- It keeps things secure.
- It always supports its customers whenever they need any kind of assistance or a resource.

Also, MySQL keeps up with database standards demanded by emerging technologies. For example, by enabling full geography support.

NoSQL + SQL = MySQL: This is an implementation of NoSQL inside the pre-existing MySQL database. It allows the use of the CRUD pattern (create, read, update and delete) for managing



data in a schemaless manner to allow developers more flexibility and not force them to learn SQL and schema management for manipulating data.

MySQL has administration capability using a single program called MySQL Shell that controls everything. Its functionalities are listed below.

- **Rapid prototyping capabilities:** With Shell you can administer your database in multiple languages such as JavaScript and Python. You can create sessions, write a small script, do rapid prototyping, and

write queries in it.

- **Full SQL and X DevAPI support:** This comes with built in auto-completion for commands.
- **InnoDB Cluster support:** You can add multiple nodes to it and set up your HA solution within minutes.
- **DevOps tool support:** If you want to stop the database, or install and uninstall plugins, Shell will help you with that.

What's in the MySQL database?

One of the key things that separates MySQL from other open source databases is that it provides the statistics about everything that is running inside the database. It tells a user about the current active connections, the CPU utilisation, memory usage, read/write operations, network bytes received/transmitted, and much more, as shown in Figure 1. All these things are in-built in the MySQL database and reside in what is called the performance schema, which runs automatically whilst keeping the data for a certain period of time. You can set it by the size that you want to assign it to in your system or the amount of time for which you

JSON functions

JSON is rapidly becoming the language of the Internet. MySQL allows native JSON objects. These can be manipulated via in-built functions that provide a very wide range of functionalities. JSON object handling is built into every part of the MySQL database such as optimiser, runtime, and replication. It is understood that there are JSON objects that are going to be queried and transferred; so we need to make sure that everything in JSON is managed optimally.